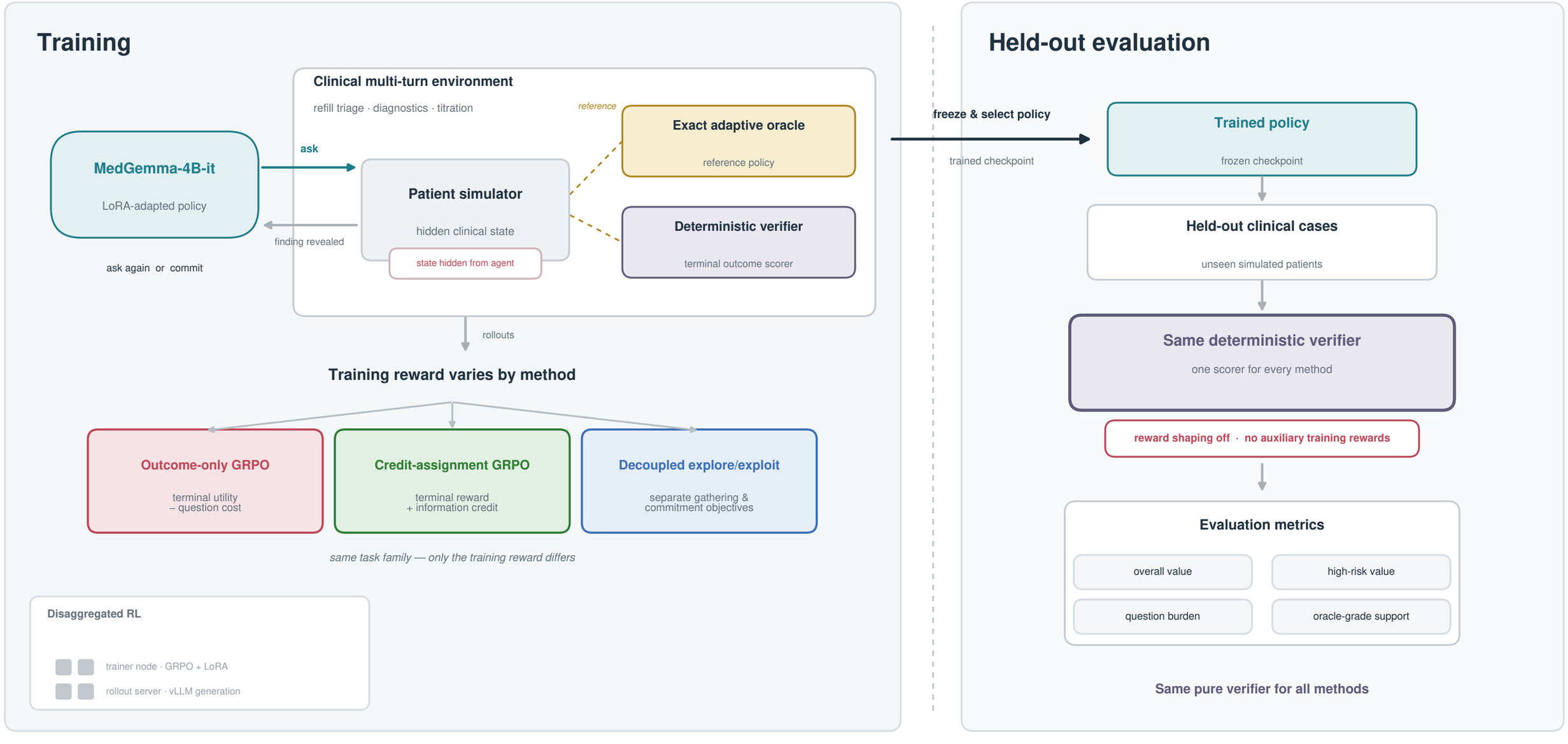
Training
Clinical multi-turn environment
refill triage · diagnostics · titration
reference
Exact adaptive oracle
reference policy
MedGemma-4B-it
LoRA-adapted policy
ask
Patient simulator
hidden clinical state
state hidden from agent
finding revealed
Deterministic verifier
terminal outcome scorer
ask again or commit
rollouts
Training reward varies by method
Outcome-only GRPO
terminal utility
– question cost
Credit-assignment GRPO
terminal reward
+ information credit
Decoupled explore/exploit
separate gathering &
commitment objectives
same task family — only the training reward differs
Disaggregated RL
trainer node · GRPO + LoRA
rollout server · vLLM generation
freeze & select policy
trained checkpoint
Held-out evaluation
Trained policy
frozen checkpoint
Held-out clinical cases
unseen simulated patients
Same deterministic verifier
one scorer for every method
reward shaping off · no auxiliary training rewards
Evaluation metrics
overall value
high-risk value
question burden
oracle-grade support
Same pure verifier for all methods